



**Stan's
Technologies**

Benchmark Report

BoolMinGeo vs PyEDA Boolean Minimization

Experiment Date: 2026-01-13

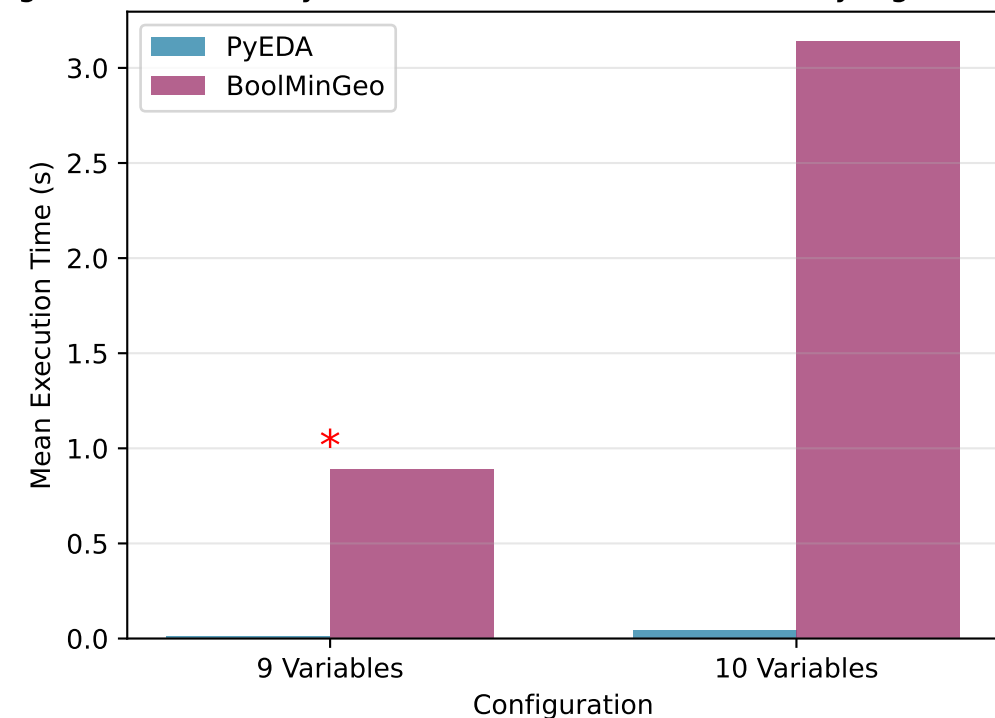
Random Seed: 42

Total Test Cases: 200

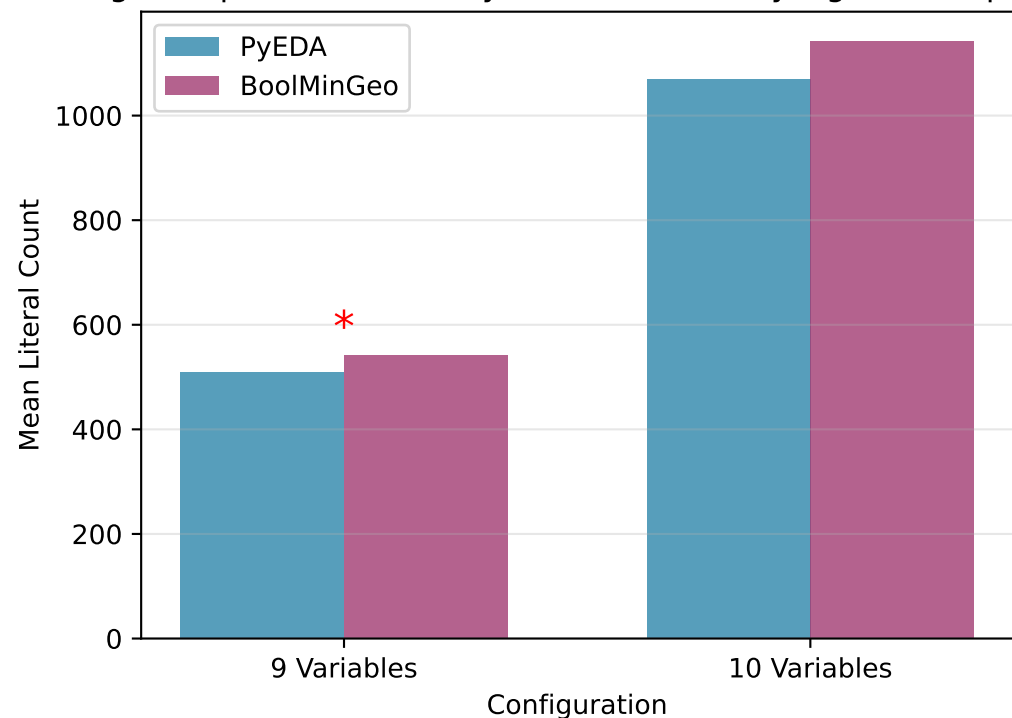
Statistical Significance Level: $\alpha = 0.05$

A Statistical Comparison of Boolean Minimization Algorithms

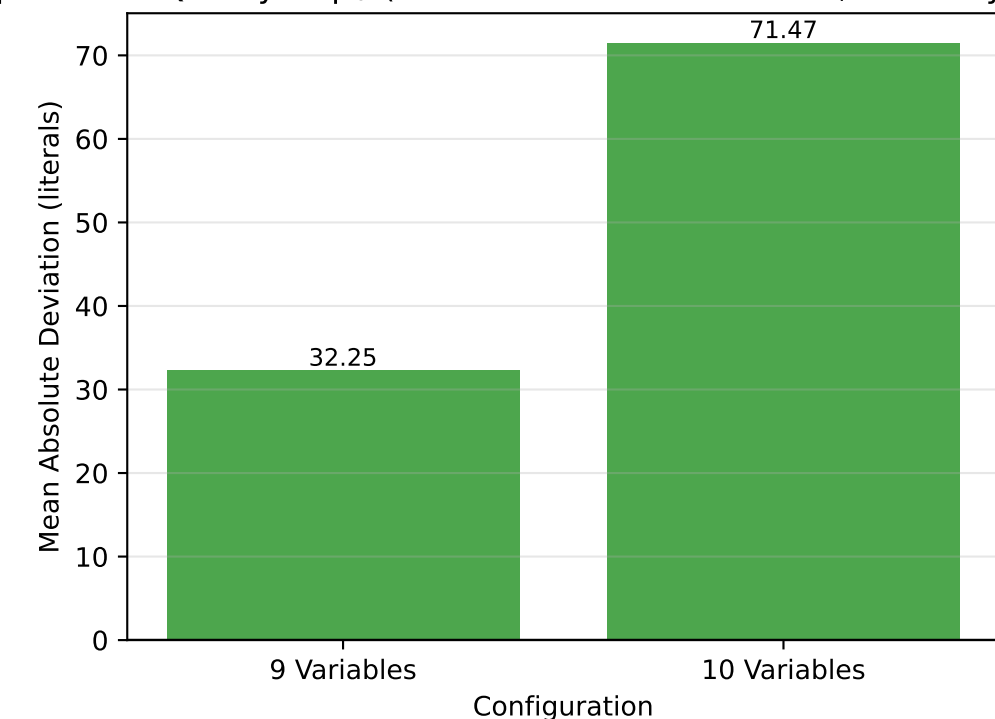
Age Performance by Variable Count\n(* = statistically significant, p < 0.05)



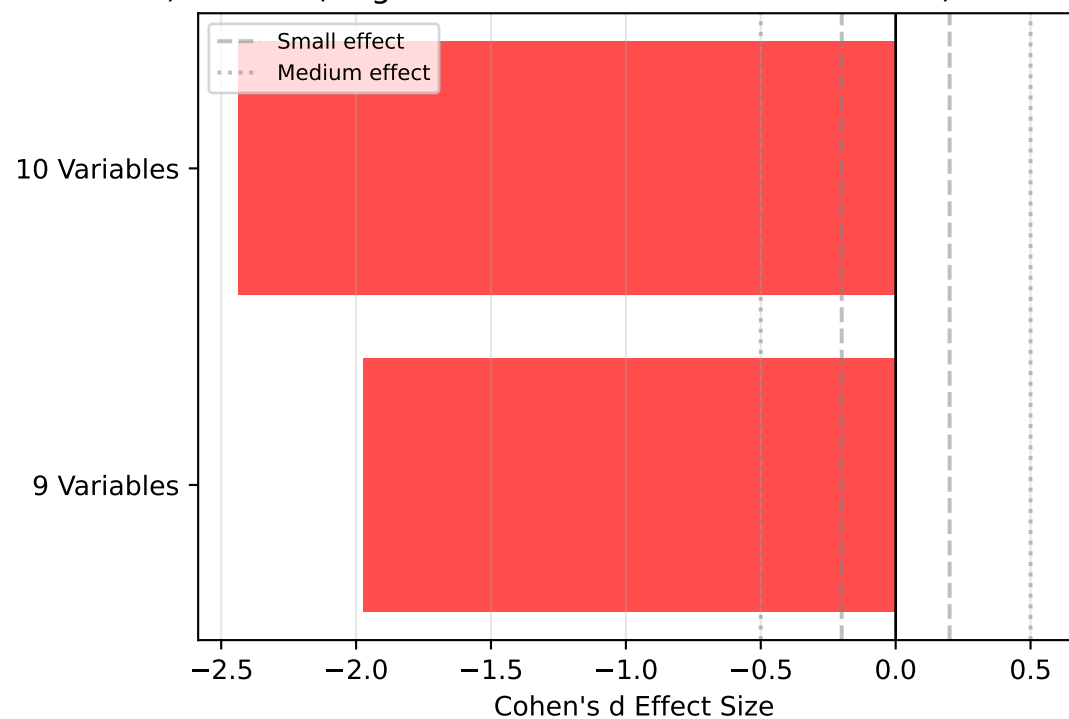
Age Simplification Quality\n(* = statistically significant, p < 0.05)



Simplification Quality Gap\n(Green = BoolMinGeo better, Red = PyEDA better)



Effect Size: Simplification Quality
(Negative = BoolMinGeo more minimal)



STATISTICAL ANALYSIS SUMMARY

EXPERIMENT OVERVIEW

=====	
Total Test Cases:	200
Configurations Tested:	2 (9-10 variables)
Tests per Config:	10 (sparse, balanced, dense distributions)
Random Seed:	42
Significance Level:	$\alpha = 0.05$

9-VARIABLE FUNCTIONS (100 tests, 100 non-constant)

EXECUTION TIME:

Mean PyEDA:	0.014598 s
Mean BoolMinGeo:	0.892981 s
Mean Difference:	-0.878383 s
95% CI:	[-0.922279, -0.834487]
p-value:	0.000000
Statistically Sig:	YES
Effect Size (d):	-3.9705 (large)
Winner:	PyEDA

SIMPLIFICATION QUALITY (Literal Count):

Mean PyEDA:	509.43 literals
Mean BoolMinGeo:	541.68 literals
Mean Difference:	-32.25 literals
Mean Deviation:	32.25 literals
Quality Gap:	6.33% relative to PyEDA
95% CI:	[-35.49, -29.01]
p-value:	0.000000
Statistically Sig:	YES
Effect Size (d):	-1.9727 (large)
Winner:	PyEDA

10-VARIABLE FUNCTIONS (100 tests, 100 non-constant)

EXECUTION TIME:

Mean PyEDA:	0.044794 s
Mean BoolMinGeo:	3.138702 s
Mean Difference:	-3.093908 s
95% CI:	[-3.235009, -2.952807]
p-value:	0.000000
Statistically Sig:	YES
Effect Size (d):	-4.3508 (large)
Winner:	PyEDA

SIMPLIFICATION QUALITY (Literal Count):

Mean PyEDA:	1070.16 literals
Mean BoolMinGeo:	1141.63 literals
Mean Difference:	-71.47 literals
Mean Deviation:	71.47 literals
Quality Gap:	6.68% relative to PyEDA
95% CI:	[-77.29, -65.65]
p-value:	0.000000
Statistically Sig:	YES
Effect Size (d):	-2.4385 (large)
Winner:	PyEDA

CONCLUSIONS

KEY FINDINGS

1. PERFORMANCE:
 - Average PyEDA time: 0.029696 s
 - Average BoolMinGeo time: 2.015842 s
 - Difference: -1.986146 s
 - Statistical significance: YES (p = 0.000000)
2. SIMPLIFICATION QUALITY:
 - Average PyEDA literals: 789.79
 - Average BoolMinGeo literals: 841.65
 - Mean deviation: 51.86 literals
 - Quality gap: 6.57% relative to PyEDA
 - Statistical significance: YES (p = 0.000000)
3. EQUIVALENCE:
 - All 200/200 tests passed logical equivalence checks
 - Both algorithms produce functionally correct results
4. EFFECT SIZES:
 - Performance: -1.6167 (large)
 - Simplification: -1.6854 (large)

INTERPRETATION

Per-Variable Analysis:

- 9 variables: 6.33% quality gap, PyEDA produces more minimal results (Mean deviation: 32.25 literals)
- 10 variables: 6.68% quality gap, PyEDA produces more minimal results (Mean deviation: 71.47 literals)

REPRODUCIBILITY

- Random seed: 42
- All test cases can be reproduced with this seed
- Statistical tests use standard methods (paired t-test, Cohen's d)

Report generated: 2026-01-13 22:01:49

© Stan's Technologies 2026